Robot cell for mounting the bearings of an LV-alternator rotor: fit physics, process window and takt

Systems-engineering dossier, revision 1.0 (10 Nov 2026). Tables marked "generated" are written from `results/studies.json` by `scripts/make_tables.py`; no result number is typed by hand. Every mounting cycle in this study is run by the compiled IEC 61131-3 controller (`plc/st/mount_cell.st`, matiec) against models of the heater, the robot, the hot slide-on and the press. The robot side is a FANUC TP program (`robot/BRG_MOUNT.ls`).

Central question. A robotised rotor line must mount the drive-end bearing (6322 C3 on a 110 mm m5 seat) and the non-drive-end bearing (6319 C3 on a 95 mm k5 seat) of a 4-pole LV alternator at the takt of a tripled output (255 s per rotor). Should the cell heat the bearings and slide them on, or press them on cold? For hot mounting, what heating, transfer, slide-on and hold times does the physics allow, and what decides the cycle?

Short answer. The cell meets the takt by heating the bearing *less*, not more, and by holding it with a clamp instead of the robot. Two different fits decide two different limits:

- the **tightest** fit (48 µm on the DE seat) sets the slide-on margin. The smallest heating that keeps a 5 µm gap even at a slow 5 mm/s slide-on is ΔT = 60 K;
- the **loosest** fit (13 µm DE, 3 µm NDE) sets the hold time: the ring has to cool back almost to the shaft temperature before it grips. At 90 K that takes 114 s (DE) and 192 s (NDE); at 60 K, 70 s and 124 s.

The usual practice of heating to about 110 °C (90 K) and holding with the robot gives 421 s per rotor, 1.65× the takt. A clamp at 90 K gives 278 s. The window minimum (60 K) with a clamp and the NDE bearing mounted first gives 196 s. If the seat diameter is gauged before mounting and the hold time is set per part from its measured interference, the cycle falls to 95 s, and it is then set by the heater. A cold servo press is faster still (72 s), but it pushes up to 64 kN through the seat, and its force window cannot tell a wrong tolerance class from friction scatter: it rejects 0.9 % of k5 shafts on the m5 seat. The DE fit itself (m5 with C3 clearance) is safe only while the inner ring runs less than about 4 K warmer than the outer ring.

How to read the evidence. Every value carries a tag: **[S]** sourced (§13), **[D]** derived here (model and file named), **[A]** assumed. The method follows Wasson, *System Engineering Analysis, Design, and Development*, 2nd ed., Wiley 2016 [W]; page numbers are the printed ones.

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0. Findings

1. **The process window has two edges from two different fits** (§6.3). The slide-on gap closes first where the interference is largest; the grip comes last where it is smallest. Raising ΔT widens the first edge and lengthens the second: from 60 to 90 K the DE slide-on margin grows from 13.6 to 33.9 µm and the loosest-fit hold from 70 to 114 s.

2. **The window minimum is 60 K** (§6.3). At 50 K the tightest DE fit still slides at 20 mm/s (6.3 µm) but seizes at 5 mm/s, and a protective stop half way through the slide-on seizes it after 3.5 s. At 60 K the margins are 13.6 / 7.1 µm (20 / 5 mm/s), and a stop may last 10.1 s. Below 50 K the DE ring seizes on the tightest fit.

3. **Transfer time is not the risk, a stop on the seat is** (§6.3). In air the 9.65 kg bearing cools slowly: at 60 K any heater-to-seat time up to 300 s keeps the margin. Once part of the ring is on the shaft, the 15–35 µm air gap conducts like a 760–1600 W/m²K contact and the ring cools in tens of seconds. The safety concept must not stop the robot in the middle of a slide-on for longer than 10 s (60 K) or 44 s (90 K).

4. **Released at its first grip, the ring opens a face gap of up to 15 µm** (§6.3; SR-01: 10 µm). The first grip is not at the face: on the tightest fit it starts 12–22 mm from the face. The ring must be held until it grips over its full width. A hold timer that is too short passes the PLC unnoticed (§9, scenario "hold 5 s").

5. **Takt at 3× output: only the clamp-held alternatives at 60 K and the press meet it** (§8): M1 421 s, M2 278 s, M3 196 s, M4 72 s, M5 95 s against 255 s. Mounting the NDE bearing first hides the longer NDE hold behind the DE mount (M2: 289 → 278 s; M3: 207 → 196 s).

6. **A per-part hold halves the hot cycle** (§8). The full-grip time falls from 70 s at 13 µm to 10 s at 48 µm (DE). Gauging the seat and setting the hold per part (M5) cuts the mean clamp wait from 135 to 34 s, and it records the one number that proves the fit: the measured interference.

7. **The press window proves the seat, not the fit** (§7). With µ = 0.10–0.20 the window at 0.9·B spans 6.2–54.5 kN; it rejects only interferences below 7.8 µm, 0.9 % of k5 shafts and 0.01 % of 5 µm undersize grinds. A

specified paste (μ 0.12–0.15) narrows it to 7.5–40.9 kN and still rejects only 1.9 % of k5 shafts. It does catch the gross faults: a missing fit, a dry seat (over-force) and an obstacle (no seat).

8. **The DE fit is limited by the operating clearance** (§6.2). m5 + C3 leaves 0.546 % of DE bearings without operating clearance at a 5 K inner/outer temperature difference; SR-04 (0.1 %) holds only up to about 3.5–4 K. CN clearance with m5 fails badly (66 % at 5 K). The NDE k5 + C3 holds to 5 K (0.057 %).

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1. Need, problem space and stakeholders ([W] ch. 4–5)

System of interest (SOI). The bearing-mounting cell of a rotor line: one 6-axis robot (25 kg class), one induction heater per bearing size, a hold-down clamp or a servo press, a seat-face sensor, and the cell PLC.

Higher-order system. The robotised rotor assembly line: shaft and rotor core → exciter armature and rotating rectifier → fan → **bearings** → balancing → rotor into the stator.

Role	Who	What they need from the SOI
Production	Rotor line	A cycle inside the takt at 3× output, with one robot
Technologist	Owner of the mounting process	A process window derived from the fit, with limits the PLC can enforce
Quality	Final test, field returns	Bearings seated against the shoulder, with operating clearance, and a record per rotor
Maintenance / safety	Cell	A cell that tolerates protective stops without scrapping rotors
End user	Generating-set operator	No bearing preload or creep: the drive-end bearing carries the rotor and the coupling

Problem statement ([W] Principle 4.15, p. 91). Manual bearing mounting heats "to about 110 °C" and pushes the bearing on by hand. A robot repeats whatever recipe it is given. The recipe parameters (heating, transfer, slide-on speed, hold time, press force window) come from the fit tolerances, the bearing's internal clearance, heat transfer and friction, and they interact.

Measures ([W] Principles 5.10–5.11, p. 115): MOE-1 cycle per rotor against the takt; MOE-2 slide-on margin (μm); MOE-3 seat (face gap after cooling); MOS-1 evidence of the fit per part; MOS-2 force through the seat; MOS-3 robustness to a protective stop.

2. Concept of operations ([W] ch. 5–6)

Phase	What happens	What the physics asks
Recipe release	Per rotor type: bearing, fit, ΔT, limits	Process window (§6.3), press window (§7)
Heat	Induction heater to shaft temperature + ΔT (never above 120 °C)	Heating time ∝ ΔT
Pick and transfer	Robot lifts the bearing and moves it to the seat start	Air cooling, slow (§6.3)
Slide-on	Robot moves the ring to the shoulder, stops on the face sensor (skip)	Gap > 5 μm everywhere until the face is at the shoulder
Hold	Robot or clamp holds the ring against the shoulder	Until full grip (§6.3)
Press (M4)	Robot leaves the ring on the lead-in; servo press strokes to the shoulder	Force window, over-force, seat rise (§7)
Record	Verdict, ΔT, transfer and hold times or the F-x points, per rotor	Traceability

UC-1 "mount a bearing": preconditions — rotor clamped, seat cleaned and gauged (M5), robot ready; trigger — the line PLC starts the cell; main scenario as above; extensions — slow transfer → back to the heater (re-heat, at most once); not seated in time → NOK; protective stop → fault; over-temperature → fault.

3. Modes and states ([W] ch. 7)

States of `MOUNT_CELL` (`plc/st/mount_cell.st`):

State	Entry	Allowed	Prohibited (Principle 7.10, p. 169)
S0 Idle	Reset	Recipe download	Heating, robot commands
S1 Heat	Start, safety OK, robot ready	Heater on	Bearing above 120 °C
S2 Pick	Bearing at shaft + ΔT	Robot picks	—
S3 Transfer	Robot reports "picked"	Robot moves	Transfer longer than the recipe limit (→ S6)

State	Entry	Allowed	Prohibited (Principle 7.10, p. 169)
S4 Insert	Robot at the seat start	Slide-on	Insert time over the limit (→ NOK, seized)
S5 Hold	Face sensor at the shoulder	Robot or clamp holds	Release before the hold time
S6 Return	Transfer too slow	Robot returns the bearing to the heater	More re-heats than allowed
S7 Press	Robot at the seat start (press methods)	Press stroke, window evaluation	Force over the press limit
S8 Done / S9 Fault	Verdict / safety, over-temperature	Reset	Any motion command

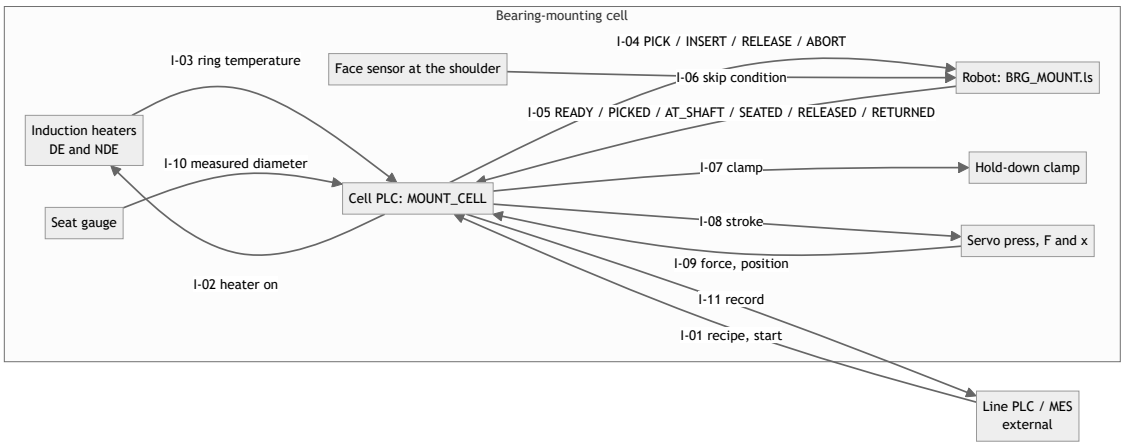
4. Requirements ([W] ch. 19–22)

ID	Requirement	Source	Verification
SR-01	The ring face shall be within 10 µm of the shaft shoulder after cooling.	[A] (a seat check after cooling)	Analysis (held until full grip); Test (feeler / dial at the face)
SR-02	While the ring slides on, the radial gap shall stay at least 5 µm everywhere, for the whole fit tolerance, at 20 mm/s and at 5 mm/s.	[A] margin for out-of-roundness and cocking	Analysis (§6.3)
SR-03	No bearing shall be heated above 120 °C.	Open bearings must not be heated above 120 °C [B1] p. 203 [S]	Analysis; PLC limit
SR-04	Fewer than 0.1 % of mounted bearings shall run without radial clearance in operation.	[A]; fit and clearance data [B1] [S]	Analysis (§6.2); type test (ring temperatures)
SR-05	The cell shall mount both bearings of a rotor within the takt of 3× today's output (255 s).	Line plan [A]: 60 → 180 rotors/day, 2 shifts of 7.5 h, OEE 0.85	Analysis (§8)
SR-06	Each mount shall leave a record that shows the seat and the fit.	Traceability [A]	Inspection
SR-07	Bearing and gripper shall be within the robot payload (25 kg).	Robot data [B5] [S]	Analysis (test)

Derived requirements ([W] Principle 21.2, p. 437):

ID	Derived requirement	From
DR-01	Heat to shaft temperature + 60 K (window minimum), not to a fixed 110 °C.	SR-02, SR-05, §6.3
DR-02	Hold the ring against the shoulder until it grips over its full width: 1.2 × the full-grip time of the loosest fit (85 s DE, 149 s NDE at 60 K), or per part from the gauged interference (M5).	SR-01, §6.3
DR-03	A protective stop during the slide-on shall not last longer than 10 s at 60 K; the cell shall then pull the ring off and re-heat it.	SR-02, §6.3
DR-04	Hold with a clamp, not with the robot; mount the NDE bearing first.	SR-05, §8
DR-05	On the DE bearing, measure the inner/outer ring temperature difference in the type test; above 3.5 K use a larger clearance group than C3.	SR-04, §6.2
DR-06	If a press is used, specify the mounting paste (µ range) and do not use the force window as evidence of the fit class.	SR-06, §7

5. Architecture and interfaces ([W] ch. 26–27)



ID	Interface	Definition
I-01	Recipe	method, ΔT, 120 °C limit, transfer limit, insert limit, hold time, press window (x1, x2, force bands), seat position
I-04/05	Robot handshake	`robot/io_map.yaml`: DI 101–104 from the PLC, DO 101–106 to the PLC; checked against the TP program and the ST variables by `tests/test_interfaces_and_plc.py`
I-06	Face sensor	Robot DI 110, SKIP CONDITION on the slide-on move (`L PR[5] R[10]mm/sec FINE Skip,LBL[80]`)
I-10	Seat gauge (M5)	Seat diameter → interference with the bearing bore class → hold time from §6.3

6. Design, budgets and margins ([W] ch. 31)

6.1 Fit physics (`mount/fits.py`)

Equal materials, solid shaft, Lamé thick-cylinder theory:

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δ = d·p/E · [ (b² + a²)/(b² − a²) + ν ] + d·p/E · (1 − ν)  =  d·p/E · 2b²/(b² − a²)      a = d/2, b = d1/2
raceway growth Δd1 = 2·u(b) = 4·p·a²·b / (E·(b² − a²))   →   Δd1/δ = a/b = d/d1          [D]
press force F(x) = μ·p·π·d·x
```

For the 6322 the raceway grows by f1 = 0.772 of the interference, for the 6319 by 0.784 [D] (the ratio a catalogue reads from a diagram [B1] p. 184). Inner-ring outer diameter d1 = d + 0.25(D − d) [A]. Interference lost to surface smoothing 0.4·(Rz_shaft + Rz_bore) = 2.6 µm [A]. Steel data [B2] [S].

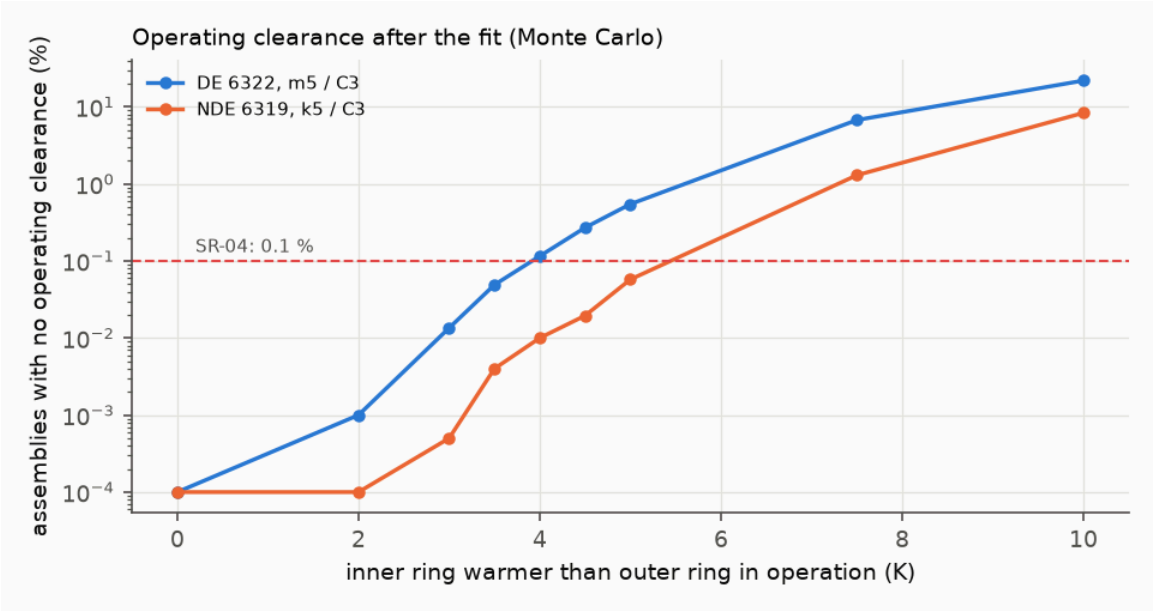
6.2 Fit selection and operating clearance (generated)

Operating clearance = catalogue clearance (uniform in its group [B1]) − f1 × effective interference − 0.012·ΔT·dm (inner/outer temperature difference, [B1] p. 184) [S formula]. Shaft and bore normally distributed in their fields (6σ = field) [A]; 200 000 draws.

Bearing	Shaft	Clearance	Interference (µm)	Max pressure (MPa)	Max hoop stress (MPa)	Cold press force, µ 0.20 (kN)	Operating clearance mean / worst case (µm)	No clearance in operation
DE	k5	C3	3–38	14.7	58	51	26.7 / -1.8	0.000 %
DE	m5	C3	13–48	18.5	73	64	19.0 / -9.5	0.546 %
DE	m6	C3	13–55	21.2	84	73	16.3 / -14.9	3.243 %
DE	n6	C3	23–65	25.1	99	87	8.5 / -22.7	21.865 %
DE	m5	CN	13–48	18.5	73	64	-4.0 / -30.5	65.528 %
DE	m6	CN	13–55	21.2	84	73	-6.7 / -35.9	75.606 %
NDE	k5	C3	3–38	16.2	68	44	21.1 / -6.6	0.057 %
NDE	m5	C3	13–48	20.5	86	55	13.3 / -14.4	6.035 %
NDE	k5	CN	3–38	16.2	68	44	1.1 / -24.6	45.234 %
NDE	m5	CN	13–48	20.5	86	55	-6.7 / -32.4	77.708 %

The catalogue's choices (m5 on the 110 mm DE seat, k5 on the 95 mm NDE seat [B1] p. 148) are right, but the DE bearing is close to the edge. The operating temperature difference is the deciding assumption:

Inner ring warmer than outer (K)	0	2	3	3.5	4	4.5	5	7.5	10
DE m5 / C3: no clearance	0.000 %	0.001 %	0.013 %	0.049 %	0.116 %	0.271 %	0.546 %	6.745 %	21.936 %
NDE k5 / C3: no clearance	0.000 %	0.000 %	0.001 %	0.004 %	0.010 %	0.019 %	0.057 %	1.298 %	8.343 %



6.3 Hot mounting: the transient model and the process window (`mount/thermal.py`)

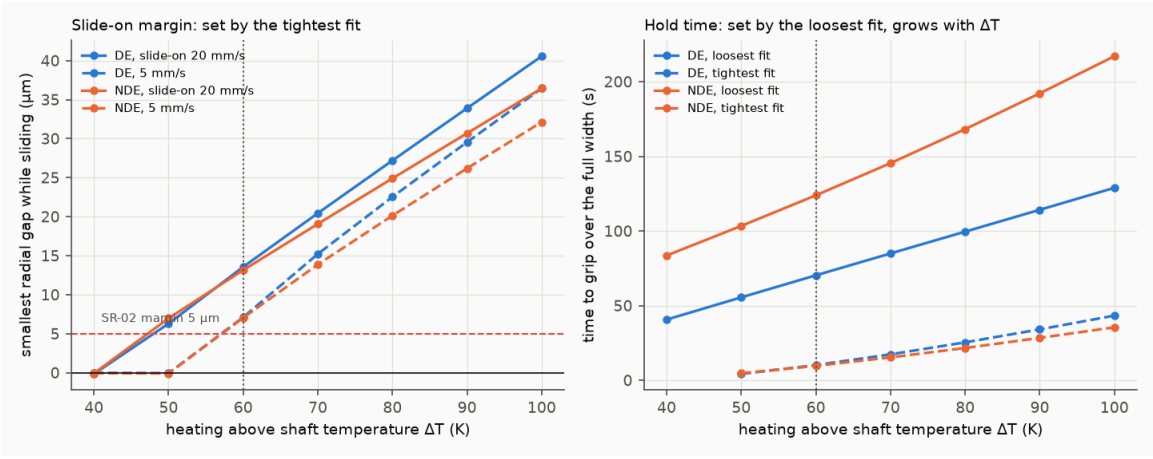
The inner ring is 20 axial nodes (radially lumped; Bi < 0.02), coupled through the balls (3 W/K [A]) to the rest of the bearing. The shaft is a 22 × 33 (r, z) finite-volume grid under and around the seat. While a ring node is over the seat it exchanges heat with the shaft surface through the radial gap:

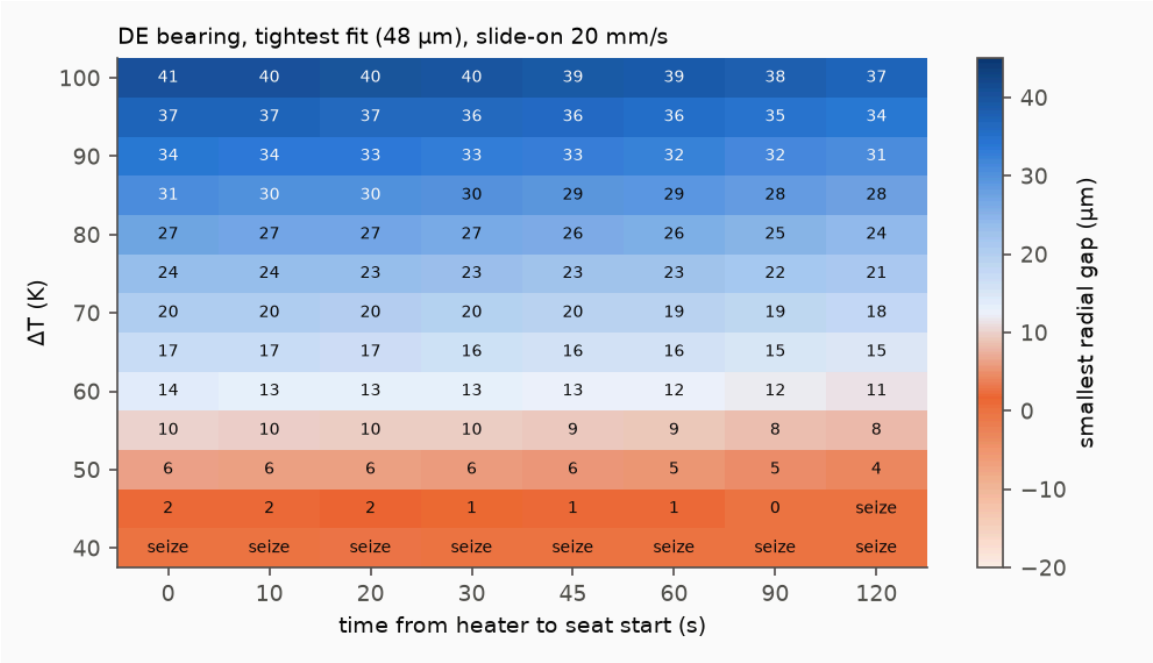
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h = k_air / (g + g_jump) + h_rad,     g = [d − δ + α·d·(T_ring − T0) − (d + α·d·(T̄_shaft − T0))] / 2
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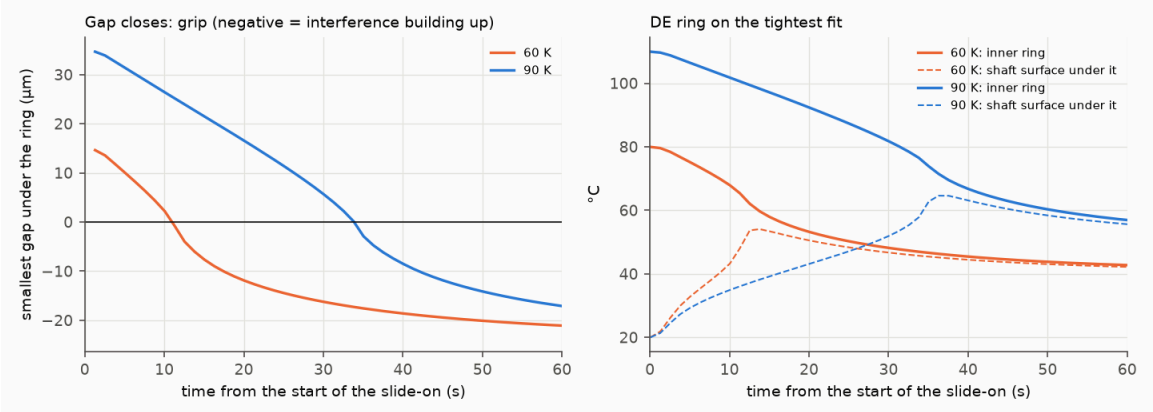
with k_air = 0.0288 W/mK at 60 °C [B3], g_jump = 3 µm [A], and 20 kW/m²K once the gap has closed [A]. T̄_shaft is the slice-wise area-mean shaft temperature: the surface displacement of a long solid cylinder with a radial profile T(r) is α·R·T̄ [D]. The leading face slides from the lead-in to the shoulder at the slide-on speed; a node whose gap closes before the face reaches the shoulder seizes the ring.

Why the gap matters more than the transfer. In air (h = 15 W/m²K [A]) the inner ring loses less than 0.1 K in the 1.9 s nominal transfer (time constant about 35 min). On the shaft, a 15–35 µm gap gives k/(g + g_jump) ≈ 760–1600 W/m²K: the ring's time constant drops from tens of minutes to tens of seconds.

Bearing	ΔT (K)	Heating (s)	Smallest gap 20 / 5 mm/s (µm)	Longest heater-to-seat time (s)	Longest stop half way (s)	Full grip, tightest / loosest fit (s)	Face gap if released at first grip, tightest / loosest (µm)
DE	40	63	-0.2 / -0.0	0	0.0	— / 40.6	— / 4.6
DE	50	79	6.3 / -0.1	76	3.5	4.2 / 55.5	3.0 / 4.6
DE	60	94	13.6 / 7.1	≥ 300	10.1	10.2 / 70.3	7.3 / 4.5
DE	70	110	20.4 / 15.2	≥ 300	19.0	17.4 / 84.9	11.0 / 5.0
DE	80	126	27.2 / 22.5	≥ 300	30.2	25.4 / 99.5	11.8 / 5.6
DE	90	142	33.9 / 29.6	≥ 300	43.6	34.1 / 114.2	14.3 / 5.0
DE	100	157	40.6 / 36.4	≥ 300	59.1	43.3 / 128.9	15.2 / 4.3
NDE	40	38	-0.1 / -0.1	0	0.0	— / 83.6	— / 1.8
NDE	50	47	7.0 / -0.1	117	4.2	4.7 / 103.4	4.7 / 1.7
NDE	60	56	13.1 / 7.0	≥ 300	9.8	9.6 / 123.9	5.1 / 1.5
NDE	70	66	19.1 / 13.9	≥ 300	17.1	15.3 / 145.4	9.3 / 1.7
NDE	80	75	24.9 / 20.1	≥ 300	26.0	21.6 / 168.2	11.4 / 2.0
NDE	90	84	30.7 / 26.2	≥ 300	36.8	28.4 / 192.2	12.2 / 2.2
NDE	100	94	36.4 / 32.1	≥ 300	48.8	35.5 / 217.1	11.4 / 2.5





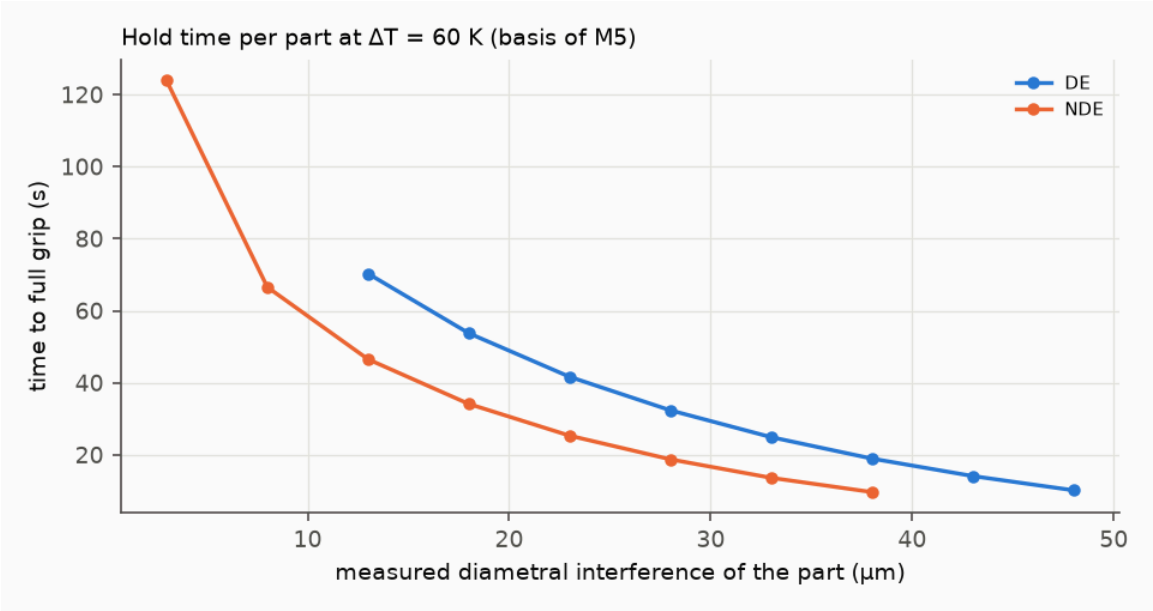


Reading the window.

- The slide-on margin grows by about 0.7 µm per K: the free expansion α·d/2 is 0.66 µm/K for the DE ring [D].
- The hold time grows with ΔT too, and it is set by the loosest fit. A ring with 13 µm of interference must cool until α·d·(T_ring − T̄_shaft) < 13 µm, about 10 K above the shaft; from 90 K that takes 114 s.
- The window minimum is the smallest ΔT whose margin holds at 5 mm/s and that tolerates a 15 s pause in transfer [A]: **60 K** for both bearings (`dT_window_min`).
- The full-grip time against the actual interference (60 K) is the basis of the per-part hold of M5:

Interference (µm)	13	18	23	28	33	38	43	48
DE full grip (s)	70.3	53.7	41.6	32.4	24.9	19.0	14.1	10.2

Interference (µm)	13	18	23	28	33	38	43	48
NDE interference (µm)	3	8	13	18	23	28	33	38
NDE full grip (s)	123.9	66.6	46.5	34.1	25.3	18.7	13.6	9.6



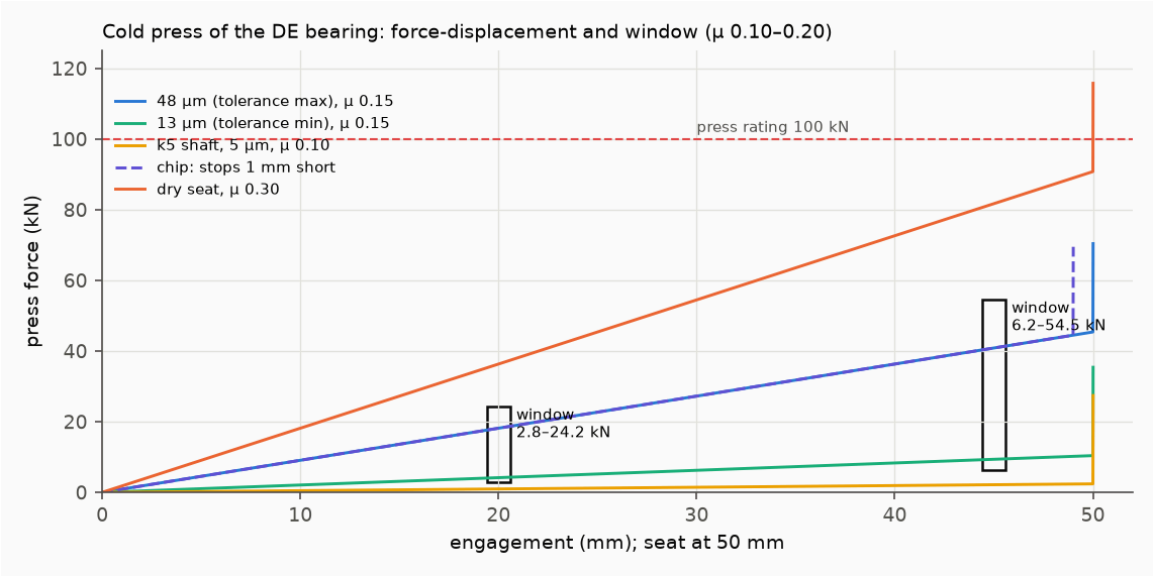
6.4 Payload and heater budget

Bearing 9.65 kg (6322) + gripper 8 kg [A] = 17.65 kg against 25 kg [B5]. Heating by 60 K: 94 s (DE) and 56 s (NDE) at 3.6 kW and 80 % [A]; by 90 K: 142 s and 84 s.

7. Cold press: the force window and what it can prove

F(x) = µ·p(δ)·π·d·x rises linearly with the engagement; at the shoulder the stroke stops and the force rises steeply (50 kN/mm [A]). The window at x1 = 0.4·B and x2 = 0.9·B is the band from (smallest effective interference, lowest µ) to (largest interference, highest µ). Because F ∝ µ·δ, the window cannot separate a low interference with high friction from a high interference with low friction. The rejected share of wrong parts is a Monte Carlo over shaft, bore and µ (20 000 draws):

Bearing	Friction range	Window at x1 (kN)	Window at x2 (kN)	Interference surely rejected below (µm)	k5 shafts rejected	5 µm undersize grind rejected	Friction work at max fit (J)
DE	µ 0.10–0.20	2.8–24.2	6.2–54.5	7.8	0.9 %	0.01 %	1600
DE	µ 0.12–0.15 (paste)	3.3–18.2	7.5–40.9	10.9	1.9 %	0.05 %	1600
NDE	µ 0.10–0.20	0.1–16.2	0.2–36.5	2.8	0.0 %	0.11 %	980
NDE	µ 0.12–0.15 (paste)	0.1–12.2	0.2–27.4	2.9	0.0 %	0.08 %	980



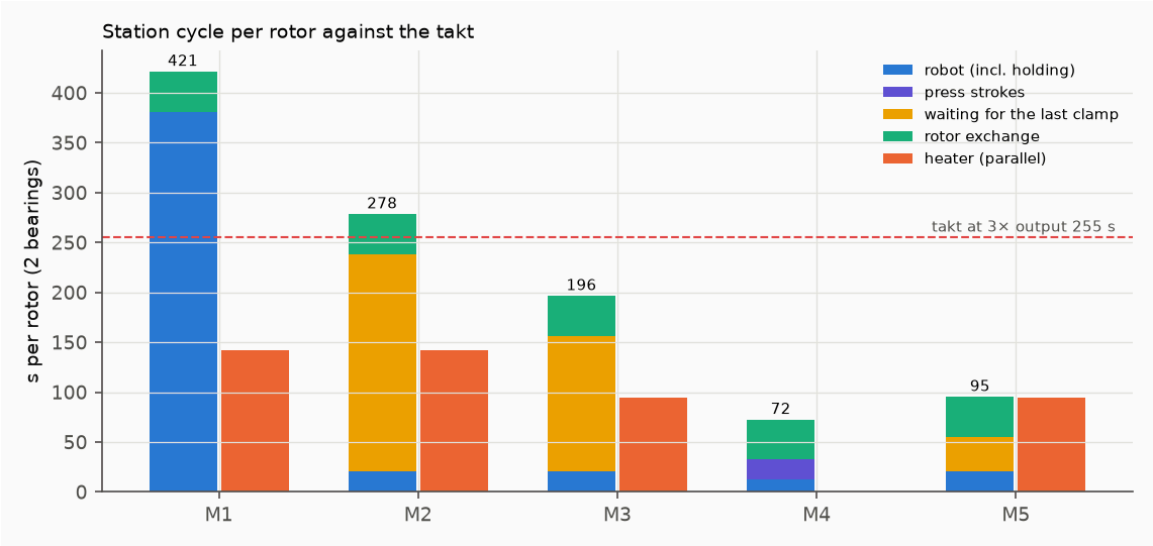
What the window does catch is the gross fault: a fit far below tolerance (k5 at 5 µm: NOK at x1), a dry seat (μ 0.30: over-force before the shoulder), and an obstacle (stroke 1 mm short: no seat rise at the seat position) (§9). The friction work dissipated at the seat is 1.6 kJ (DE) at the largest fit and μ 0.20 [D], the reason hot mounting is preferred for large bearings when the seat surface matters.

8. Trade study: analysis of alternatives ([W] ch. 32)

	Method	Heating	Hold	Fit evidence
M1	Hot mount	90 K	Robot holds	ΔT and times only
M2	Hot mount	90 K	Clamp; robot released	ΔT and times only
M3	Hot mount	Window minimum (60 K)	Clamp	ΔT and times only
M4	Cold servo press	—	—	F-x window (seat, gross faults)
M5	Hot mount, per-part hold	60 K	Clamp, hold from gauged interference	Measured interference per part

Each alternative was run in the closed loop for both bearings at both tolerance ends:

	Alternative	ΔT (K)	Robot (s)	Press (s)	Wait for clamp (s)	Heater (s)	Best order	Cycle per rotor (s)	Takt 255 s
M1	Hot mount 90 K, robot holds until locked	90	381	0	0	142	DE-NDE	421	**misses**
M2	Hot mount 90 K, hold-down clamp, robot released	90	21	0	217	142	NDE-DE	278	**misses**
M3	Hot mount at the window minimum, hold-down clamp	60	21	0	135	94	NDE-DE	196	meets
M4	Cold servo press with force-displacement window	0	13	20	0	0	DE-NDE	72	meets
M5	As M3, hold time per part from the gauged seat interference	60	21	0	34	94	NDE-DE	95	meets



Requirement	M1	M2	M3	M4	M5
SR-01 seat	pass	pass	pass	pass	pass
SR-02 no seizure	pass	pass	pass	pass	pass
SR-03 120 °C	pass	pass	pass	pass	pass
SR-05 takt	**fail**	**fail**	pass	pass	pass

SR-04 (clearance) is a fit choice, the same for every method (§6.2); SR-06 (record) is met by all, with the different evidence listed above; SR-07 is met (§6.4).

Trade study report (Principle 32.9, p. 698): recommendation.

1. **M5**: hot mounting at the window minimum (60 K), clamp hold, NDE first, hold time per part from a gauged seat diameter. 95 s per rotor (heater-bound), no press force through the seat, and the only alternative that records the actual interference of every bearing. It needs a seat gauge and a recipe calculation in the PLC (DR-02).
2. **M3** if no gauge is fitted: 196 s, inside the takt with 59 s to spare.
3. **M4** only if the seat surface and up to 64 kN at the shoulder are acceptable; specify the paste (DR-06) and do not read its window as fit evidence.
4. **Not M1**: the robot-held hot mount at 90 K is the manual practice automated, and it needs 421 s.

9. Reliability: FMEA ([W] ch. 34)

S, O, D rated 1–10 [A]; RPN = S·O·D. Items marked ▲ are quantified in the closed loop.

ID	Failure mode	Effect	Detection	S	O	D	RPN	Mitigation
FM-01 ▲	Protective stop half way through the slide-on	Ring grips part-way: seized, rotor to rework	Insert time-out (NOK "not seated")	8	3	2	48	DR-03: stop ≤ 10 s at 60 K, else pull off and re-heat; mute non-safety stops during the 2.6 s slide-on
FM-02 ▲	Robot slow or paused in transfer	Ring cooler at the seat	Transfer timer	5	3	2	30	Return to the heater (re-heat once)
FM-03 ▲	Hold timer shorter than the full grip	Face gap up to 15 µm after cooling	**None in the PLC**	7	2	9	126	DR-02 hold from the model; face-gap check after cooling on a sample basis
FM-04 ▲	Shaft from the wrong tolerance class (press)	Wrong clearance / creep	F-x window catches only gross cases	7	2	8	112	Seat gauge (M5)
FM-05 ▲	Dry seat (press)	Over-force, seat scoring	Over-force	6	3	2	36	Paste dosing check
FM-06 ▲	Chip or cocked ring (press)	Ring not at the shoulder	No seat rise	8	3	2	48	Seat cleaning, NOK
FM-07	Heater temperature probe drifts high	Bearing above 120 °C	Second probe / plausibility vs heating time	7	2	5	70	Heating-time plausibility (energy balance, §6.4)
FM-08	Bearing heated without the shaft temperature known	ΔT wrong	Shaft probe missing	5	3	3	45	Shaft temperature is a recipe input

ID	Failure mode	Effect	Detection	S	O	D	RPN	Mitigation
FM-09	Inner/outer ring difference > 4 K in service	Bearing preloaded	Bearing temperature in the type test	8	3	6	144	DR-05

Scenarios run in the closed loop (generated, DE bearing):

Scenario (DE bearing)	PLC verdict	Reason	Re-heats	Smallest gap (µm)	Held until full grip
Protective stop of 10 s half way, 90 K (M2)	OK	—	0	25.8	yes
Slide-on at 5 mm/s, window minimum (M3)	OK	—	0	7.1	yes
Protective stop of 10 s half way, window minimum (M3)	OK	—	0	0.5	yes
Robot paused 90 s in transfer, 50 K, transfer interlock on	OK	—	1	6.3	yes
Same, interlock off	OK	—	0	4.7	yes
Hold 5 s instead of the recipe, loosest fit (M2)	OK	—	0	51.9	**no**
Press: k5 shaft at 5 µm, µ 0.10 (M4)	NOK	force at x1	0	—	—
Press: dry seat µ 0.30, 48 µm (M4)	NOK	over-force	0	—	—
Press: chip, stroke stops 1 mm short (M4)	NOK	no seat	0	—	—

The "hold 5 s" row is the important one: the PLC sees a normal cycle, and only the model says the ring was released 100 s before it had gripped. That is why the hold time is a derived requirement (DR-02) and not a tuning value.

10. Verification and validation ([W] ch. 13, 28)

10.1 Verification matrix

Req.	Method here	Evidence	In the plant
SR-01	Analysis (closed loop, both tolerance ends)	§8, `seat_held_until_full_grip`	Face gap after cooling (feeler gauge), first 50 rotors
SR-02	Analysis	§6.3	Thermocouple on the inner ring; measured slide-on force on the robot (force sensor)
SR-03	Analysis + PLC limit	§3, test	Heater log
SR-04	Analysis (Monte Carlo)	§6.2	Ring temperatures in the type test
SR-05	Analysis	§8	Cycle measurement after commissioning
SR-07	Analysis	§6.4	Robot load check (payload identification)

10.2 Model verification (each against an independent result; `tests/`)

Check	Independent reference	Result
Tolerance stack	Catalogue limits [B1]	13–48 µm
Lamé pressure	General two-material press-fit formula	1e-12
Raceway growth f1 = a/b	Lamé displacement field of the ring alone	1e-9
Press force	Linear in engagement, zero without interference	exact
Clearance Monte Carlo	Worst-case formula with zero spread	1e-6
First gap on the seat	Free thermal expansion (α·d·ΔT − δ)/2	within 0.3 µm
Transfer cooling	Newton cooling (balls and radiation off)	within 0.3 K
Hotter ring: more margin, later grip	Physics of §6.3	pass
Too cold ring seizes	Physics of §6.3	pass
Heater energy balance	m·c·ΔT	exact
Robot move time	Trapezoidal / triangular profile	exact
Robot program ↔ I/O map ↔ ST variables	Static check	pass
Compiled ST: hold time, re-heat, seizure, press window, safety stop, over-temperature	Specified behaviour (§3)	pass

10.3 Validation

On a real cell: one instrumented DE inner ring (two thermocouples) and a dial at the face, mounted at 60 K on a shaft gauged at both ends of the tolerance; the measured grip time (the ring can no longer be turned by hand on

the gripper) against §6.3, and the face gap after cooling against SR-01.

11. Assumed versus sourced, and limits

Sourced [S]: bearing dimensions, masses, ratings [B1]; shaft tolerance classes and limit deviations, bore tolerance, internal clearance groups, the clearance-loss formula, the 120 °C limit [B1]; steel properties [B2]; air conductivity [B3]; greased steel friction [B4]; robot payload, reach and repeatability [B5].

Assumed [A], and where it matters:

- **Inner/outer ring temperature difference (5 K).** Decides SR-04 for the DE bearing (§6.2).
- **Gap jump distance, contact conductance, ball conductance.** They set the grip times; the ranking of the alternatives depends on the hold being long compared with the robot work, which holds for any value that keeps the grip above 10 s.
- **μ range of the press.** Sets the width of the window (§7).
- **Smoothing 0.4·(Rz1 + Rz2).** Shifts every effective interference by 2.6 μm.
- **Layout, speeds, demand.** Set the takt and the robot time.

Limits: axisymmetric, no cocking of the ring; the shaft surface displacement uses the slice mean temperature (no axial coupling of the displacement); the outer ring and balls are one node; friction during the hot slide-on is not modelled (the gap is open until it closes).

12. Decision log

#	Decision	Alternatives	Rationale
D1	Transient (ring nodes + shaft FV) model of the slide-on	Static "ΔT gives clearance" check	The static check says 40 K is enough for the DE fit (52.8 μm of expansion > 48 μm); the transient model shows it seizes, because the ring cools on the seat
D2	Hold until full grip, not first grip	Release at first grip	First grip is 12–22 mm from the face; releasing then opens up to 15 μm (§6.3)
D3	Window minimum includes a 5 mm/s slide-on	20 mm/s only	Force-controlled approaches run slowly; 50 K passes at 20 mm/s and seizes at 5 mm/s
D4	Warm-assisted press (40 K + press) dropped	Keep as M6	The transient model shows the ring gripping part-way at 40 K, so the press would push against a ring whose interference grows during the stroke: neither a clean hot mount nor an interpretable F-x curve
D5	Run the compiled ST controller in every cycle	Python re-implementation	One source of truth; the robot-PLC handshake is tested against the real variable names
D6	Mount NDE first	DE first	Hides the longer NDE grip behind the DE mount (§8)
D7	One heater per bearing size	One heater for both	Keeps heating off the critical path (95 s cycle is heater-bound even so)

13. References

- **[W]** C. S. Wasson, *System Engineering Analysis, Design, and Development*, 2nd ed., Wiley, 2016.
- **[B1]** SKF, *Rolling bearings* catalogue (PDF): p. 38 Table 2 (bore tolerance, normal class), p. 148 Table 5 (shaft fits), pp. 160–162 Tables 14–15 (ISO 286 limit deviations), p. 184 (clearance reduction by fit and temperature), p. 203 (hot mounting, 120 °C), p. 252 Table 6 (radial internal clearance), pp. 290, 292 (6319, 6322). https://cdn.skfmediahub.skf.com/api/public/0901d196802809de/pdf_preview_medium/0901d196802809de_pdf_preview_n
- **[B2]** 100Cr6 bearing steel data (density, E, ν, α, cp, k): https://steelnavigator.ovako.com/steel-grades/100cr6/
- **[B3]** Air thermal conductivity vs temperature: https://www.engineeringtoolbox.com/air-properties-viscosity-conductivity-heat-capacity-d_1509.html
- **[B4]** Friction coefficients (steel on steel, greased, static 0.16): https://www.engineeringtoolbox.com/friction-coefficients-d_778.html
- **[B5]** FANUC M-20iD/25 specifications (payload 25 kg, reach 1831 mm, repeatability ±0.02 mm): https://www.fanucamerica.com/products/robots/series/m-20/m-20id-25
- **[B6]** ANSI/EASA AR100-2025, Table 2-6: shaft fits for bearings no. 19 (k5) and no. 22 (m5), adapted from ABMA 7 (cross-check of [B1]).